Hospital Management System

1) Patients

Purpose: Store core identity and contact/insurance info for each patient.

Columns

Column	Type	Constraints
`P_id`	INTEGER	**PK**, NOT NULL
`P_name`	VARCHAR(100)	NOT NULL
`P_dob`	DATE	NOT NULL
`P_contact`	VARCHAR(255)	NOT NULL
`P_insurance`	VARCHAR(255)	NULL

2) Doctor

Purpose: Stores doctor identity and qualifications.

Columns

Column	Type	Constraints
`D_id`	INTEGER	**PK**, NOT NULL
`D_name`	CHAR(100)	NOT NULL
`D_special`	CHAR(100)	NOT NULL
`D_contact`	VARCHAR(255)	NOT NULL
`D_licence`	VARCHAR(255)	NULL/UNIQUE

3) Appointment

Purpose: Capture each scheduled encounter between a patient and a doctor.

Columns

Column	Type	Constraints
`A_id`	INTEGER	**PK**, NOT NULL
`A_date`	DATE	NOT NULL
`A_time`	TIME	NOT NULL
`A_reason`	VARCHAR(255)	NOT NULL
`A_status`	VARCHAR(255)	NOT NULL
`P_id`	INTEGER	**FK → Patients(P_id)**, NOT NULL
`D_id`	INTEGER	**FK → Doctor(D_id)**, NOT NULL

4) Ward

Purpose: Define hospital wards and basic capacity/availability.

Columns

Column	Type	Constraints
`W_id`	INTEGER	**PK**, NOT NULL
`W_name`	VARCHAR(255)	NOT NULL
`W_capacity`	INTEGER	NOT NULL, CHECK ≥ 0
`W_type`	CHAR(1)	NOT NULL
`W_available_bed`	INTEGER	NOT NULL, CHECK ≥ 0
`P_id`	INTEGER	**FK → Patients(P_id)**, NULL
`D_id`	INTEGER	**FK → Doctor(D_id)**, NULL

5) Treatment

Purpose: Catalog treatments given to patients and costing.

Columns

Column	Type	Constraints
`T_id`	INTEGER	**PK**, NOT NULL
`T_desc`	VARCHAR(255)	NOT NULL
`cost`	DECIMAL(10,2)	NOT NULL, CHECK ≥ 0

duration	INTEGER	NOT NULL
bed_rate	INTEGER	NOT NULL
P_id	INTEGER	**FK → Patients(P_id)**, NOT NULL

6) Treatment_order

Purpose: Bridge between **Appointment** and **Treatment** (what was ordered in that visit).

Columns

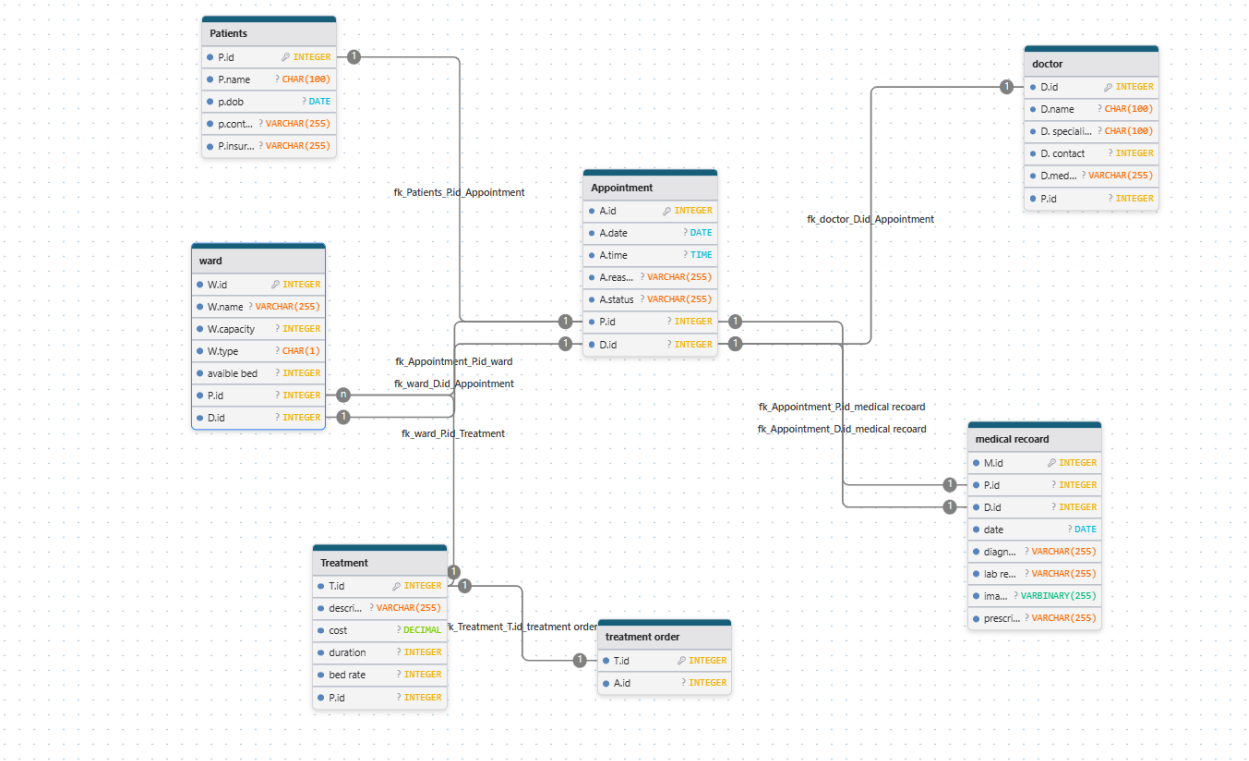
Column	Type	Constraints
T_id	INTEGER	**FK → Treatment(T_id)**, NOT NULL
A_id	INTEGER	**FK → Appointment(A_id)**, NOT NULL

7) Medical_record

Purpose: Clinical documentation generated from appointments by doctors for patients.

Columns

Column	Type	Constraints
MR_id	INTEGER	**PK**, NOT NULL
P_id	INTEGER	**FK → Patients(P_id)**, NOT NULL
D_id	INTEGER	**FK → Doctor(D_id)**, NOT NULL
A_id	INTEGER	**FK → Appointment(A_id)**, NULL/NOT NULL
MR_date	DATE	NOT NULL
MR_diag	VARCHAR(255)	NOT NULL
MR_notes	VARCHAR(255)	NOT NULL
MR_presc	VARCHAR(255)	NOT NULL

Relationship Table

Relationship	Cardinality
Patient ↔ Appointment	1 : M
Doctor ↔ Appointment	1 : M
Appointment ↔ Treatment	1 : M (per appointment, many treatments)
Appointment ↔ Medical_record	1 : M
Patient ↔ Medical_record	1 : M
Doctor ↔ Medical_record	1 : M
Patient ↔ Ward (Admissions over time)	M : M (with time)
Doctor ↔ Ward (assignment/rota)	M : M (with time)
Appointment ↔ Appointment (follow-up)	1 : M (self)
Appointment ↔ Doctor (referral)	1 : M (to specialist)